

CENTRAL LIMIT THEOREM

Central Limit Theorem: If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2 , then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}},$$

as $n \rightarrow \infty$, is the standard normal distribution $n(z; 0, 1)$.

An electrical firm manufactures light bulbs that have a length of life that is approximately **normally distributed**, with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that **a random sample** of 16 bulbs will have an average life of less than 775 hours

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}},$$

Given that, $\mu=800$, $\sigma=40$, $n=16$, then

$$P(x < 775) = P(Z < \frac{775-800}{\frac{40}{\sqrt{16}}})$$

$$= P(Z < -2.5) = 0.5 - P(0 < Z < 2.5)$$

$$= 0.5 - 0.4938 = 0.0062$$

If we have two independent populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 and if $\bar{X}_1$ and $\bar{X}_2$ are the sample means of two independent random samples of sizes n_1 and n_2 from these populations, then the sampling distribution of

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \quad (7-9)$$

is approximately standard normal, if the conditions of the central limit theorem apply. If the two populations are normal, the sampling distribution of Z is exactly standard normal.

The effective life of a component used in a jet-turbine aircraft engine is a random variable with mean 5000 hours and standard deviation 40 hours. The distribution of effective life is fairly close to a normal distribution. The engine manufacturer introduces an improvement into the manufacturing process for this component that increases the mean life to 5050 hours and decreases the standard deviation to 30 hours. Suppose that a random sample of $n_1 = 16$ components is selected from the “old” process and a random sample of $n_2 = 25$ components is selected from the “improved” process. What is the probability that the difference in the two sample means is at least 25 hours? Assume that the old and improved processes can be regarded as independent populations.

To solve this problem, we first note that the distribution of $\bar{X}_1$ is normal with mean $\mu_1 = 5000$ hours and standard deviation $\sigma_1/\sqrt{n_1} = 40/\sqrt{16} = 10$ hours, and the distribution of $\bar{X}_2$ is normal with mean $\mu_2 = 5050$ hours and standard deviation $\sigma_2/\sqrt{n_2} = 30/\sqrt{25} = 6$ hours. Now the distribution of $\bar{X}_2 - \bar{X}_1$ is normal with mean $\mu_2 - \mu_1 = 5050 - 5000 = 50$ hours and variance $\sigma_2^2/n_2 + \sigma_1^2/n_1 = (6)^2 + (10)^2 = 136$ hours². This sampling distribu-

Corresponding to the value $\bar{x}_2 - \bar{x}_1 = 25$

$$z = \frac{25 - 50}{\sqrt{136}} = -2.14$$

$$\begin{aligned} P(\bar{X}_2 - \bar{X}_1 \geq 25) &= P(Z \geq -2.14) \\ &= 0.9838 \end{aligned}$$

Q.

A synthetic fiber used in manufacturing carpet has tensile strength that is normally distributed with mean 75.5 psi and standard deviation 3.5 psi. Find the probability that a random sample of $n = 6$ fiber specimens will have sample mean tensile strength that exceeds 75.75 psi.

$$\mu_{\bar{X}} = 75.5 \text{ psi}, \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{3.5}{\sqrt{6}} = 1.429$$

$$P(\bar{X} \geq 75.75) = P\left(\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \geq \frac{75.75 - 75.5}{1.429}\right)$$

$$= P(Z \geq 0.175) = 1 - P(Z \leq 1.75)$$

$$= 1 - 0.56945 = 0.43055$$

QUESTION

PVC pipe is manufactured with a mean diameter of 1.01 inch and a standard deviation of 0.003 inch. Find the probability that a random sample of $n = 9$ sections of pipe will have a sample mean diameter greater than 1.009 inch and less than 1.012 inch.

$$\begin{aligned} P(1.009 \leq \bar{X} \leq 1.012) &= P\left(\frac{1.009 - 1.01}{0.003 / \sqrt{9}} \leq \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \leq \frac{1.012 - 1.01}{0.003 / \sqrt{9}}\right) \\ &= P(-1 \leq Z \leq 2) = P(Z \leq 2) - P(Z \leq -1) \\ &= 0.9772 - 0.1587 = 0.8385 \end{aligned}$$

Assignment

QUESTION

A random sample of size $n_1 = 16$ is selected from a normal population with a mean of 75 and a standard deviation of 8. A second random sample of size $n_2 = 9$ is taken from another normal population with mean 70 and standard deviation 12. Let $\bar{X}_1$ and $\bar{X}_2$ be the two sample means. Find

(a) The probability that exceeds 4

The probability that $3.5 \leq \bar{X}_1 - \bar{X}_2 \leq 5.5$

